

Nithin Govindarajan

Scientific Computing | Numerical Linear Algebra | Solver Development

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Areas of expertise

Numerical software development, numerical linear algebra, numerical analysis, PDE solvers, dynamical systems and control, signal processing, computational algebraic geometry, tensor methods.

Education

- 2014–2018 PhD in Mechanical Engineering, University of California in Santa Barbara
Dissertation: Periodic approximations and spectral analysis of the Koopman operator.
- 2009–2012 MSC in Aerospace Engineering (with distinction), Technische Universiteit Delft
Thesis: Optimal control approach for estimating aircraft command margins.
- 2006–2009 BSC in Aerospace Engineering (with distinction), Technische Universiteit Delft

Work experience

- 2023–present Founder, VGN Computational Consultancy Services CommV.
- 2018–present Co-founder and CTO, Labmatisse B.V., The Netherlands.
- 2026–present Researcher, KU Leuven, Belgium
- 2019–2026 Postdoctoral researcher, KU Leuven, Belgium
- 2019 Lecturer Mathematics, University of Amsterdam (UvA), The Netherlands
- 2012–2013 Junior R&D engineer, National Aerospace Laboratory (NLR), Amsterdam, The Netherlands
- 2011–2012 M.Sc. thesis Intern at NASA Ames Research Center, Mission Critical Technologies Inc., Moffett Field, CA

Software development

- 2026–present Developing global solvers for nonlinear boundary value problems using computational algebraic geometry and numerical linear algebra.
- 2018–present Developing the core algorithm engine for [Matisse.ai](#), an AI-assisted dental application, using inverse modelling and numerical optimization.
- 2022–2025 One of the main developers of [PyTensorlab](#), a Python package for tensor computations.
- 2024 Developed [SSSMatrices](#), a Julia package for sequentially semi-separable matrix computations.
- 2014–2026 Developed research code for numerical PDEs, dynamical systems, tensor decompositions, Macaulay nullspace computations, and low-rank structured linear algebra in Matlab, Python, and Julia; selected code and details are available on [my website](#).

Fellowships, honors & awards

- 2016 CCDC fellowship, Center for Control, Dynamics and Computation, Santa Barbara, CA
- 2014 Department Merit Fellowship (UCSB Mech. Eng), Santa Barbara, CA
- 2011 Huygens Scholarship Programme, Nuffic, The Hague, The Netherlands

Publications

Journals

- 2025 Govindarajan, N., Chandrasekaran, S., De Lathauwer, L. Tree quasi-separable matrices: A simultaneous generalization of sequentially and hierarchically semiseparable representations. *SIAM J. Matrix Anal. Appl.*, 46(2), 1562-1586.
- 2024 Govindarajan, N., Widdershoven, R., Chandrasekaran, S., & De Lathauwer, L. A fast algorithm for computing Macaulay nullspaces of bivariate polynomial systems. *SIAM J. Matrix Anal. Appl.*, 45(1), 368-396.
- 2022 Govindarajan, N., Vervliet, N., & De Lathauwer, L. Regression and classification with spline-based separable expansions. *Frontiers Big Data*, 5, 688496.
- 2022 Govindarajan, N., Epperly, E. N., & De Lathauwer, L. $(L_r, L_r, 1)$ -decompositions, sparse component analysis, and blind separation of sums of exponentials. *SIAM J. Matrix Anal. Appl.*, 43(2), 912-938.
- 2021 Epperly, E. N., Govindarajan, N., & Chandrasekaran, S. Minimal rank completions for overlapping blocks. *Linear Algebra Appl.*, 627, 185-198.
- 2021 Govindarajan, N., Mohr, R., Chandrasekaran, S., & Mezic, I. On the approximation of Koopman spectra of measure-preserving flows. *SIAM J. Appl. Dyn. Syst.*, 20(1), 232-261.
- 2019 Govindarajan, N., Mohr, R., Chandrasekaran, S., & Mezic, I. On the approximation of Koopman spectra for measure-preserving transformations. *SIAM J. Appl. Dyn. Syst.*, 18(3), 1454-1497.

- 2015 Govindarajan, N., De Visser, C. C., Van Kampen, E., Krishnakumar, K., Barlow, J., & Stepanyan, V. Optimal control framework for estimating autopilot safety margins. *J. Guidance Control Dyn.*, 38(7), 1197-1207.
- 2014 Govindarajan, N., de Visser, C. C., & Krishnakumar, K. A sparse collocation method for solving time-dependent HJB equations using multivariate B-splines. *Automatica*, 50(9), 2234-2244.

Conference Proceedings

- 2026 Madala, V. C., Govindarajan, N., & Chandrasekaran, S. MNO: A Multi-modal Neural Operator for Parametric Nonlinear BVPs. *AI2ASE*, Singapore.
- 2023 Widdershoven, R., Govindarajan, N., De Lathauwer, L. Overdetermined systems of polynomial equations: tensor-based solution and application. *EUSIPCO*, Helsinki.
- 2018 Chandrasekaran, S., Govindarajan, N., & Rajagopal, A. Fast algorithms for displacement and low-rank structured matrices. *ISSAC*, 17-22.
- 2016 Govindarajan, N., Arbabi, H., Van Bargaan, L., Matchen, T., & Tegling, E. An operator-theoretic viewpoint to non-smooth dynamical systems: Koopman analysis of a hybrid pendulum. *IEEE CDC*, 6477-6484.

Preprints & Technical Reports

- 2026 Govindarajan, N., Vervliet, N., Widdershoven, R., De Lathauwer, L. Efficient Krylov methods for overdetermined polynomial systems. Technical Report 26-108, ESAT-STADIUS, KU Leuven.
- 2025 Widdershoven, R., Govindarajan, N., De Lathauwer, L. Fast Macaulay null space through the intersection of shifted null spaces. Technical Report 25-56, ESAT-STADIUS, KU Leuven.
- 2019 Chandrasekaran, S., Epperly, E. N., Govindarajan, N. Graph-induced rank structures and their representations. arXiv:1911.05858.

Conferences

- 2025 Fast Macaulay Null Space, LMRA 25; Towards efficient algorithms for approximately solving overdetermined polynomial systems, Householder Symposium XXII (poster).
- 2024 Rank structured matrices induced by dynamical systems on graphs, NUMA seminar; Macaulay matrices, low displacement rank, and efficient null-space computation, Back to the Roots.
- 2023 A tensor-based approach to solving systems of multivariate polynomials, CAM23; Efficient Computation of Macaulay Matrix Null Spaces Through Exploiting Shift-Invariant Structures, SIAM AG23.
- 2022 A fast algorithm for computing Macaulay null spaces of bivariate polynomial systems, Workshop on Solving Polynomial Equations and Applications, CWI Amsterdam (poster).
- 2021 $(L_r, L_r, 1)$ -decompositions, sparse component analysis, and blind separation of sums of exponentials, SeLMA; Spline-based separable expansions for approximation, regression and classification, IPAM Workshop I.
- 2017 A toolbox for computing spectral properties of dynamical systems, SIAM DS17.
- 2016 An operator-theoretic viewpoint to non-smooth dynamical systems: Koopman analysis of a hybrid pendulum, IEEE CDC.

Teaching

- 2019–20 Lead instructor: Numerical mathematics, Amsterdam University College.
- 2019–22 Co-instructor: Fast algorithms for dense structured matrices; Numerieke modellering & benadering, KU Leuven.
- 2019–22 Teaching assistant: Numerieke wiskunde, KU Leuven.
- 2014–18 Teaching assistant: control theory, electrical circuits lab, Matlab programming, dynamics, physics lab, vibrations, and statics, UCSB.

Software skills

Python, Matlab, Julia, Git, LaTeX, Mathematica; working knowledge of C++.

Languages

English, Dutch, Tamil (basic).